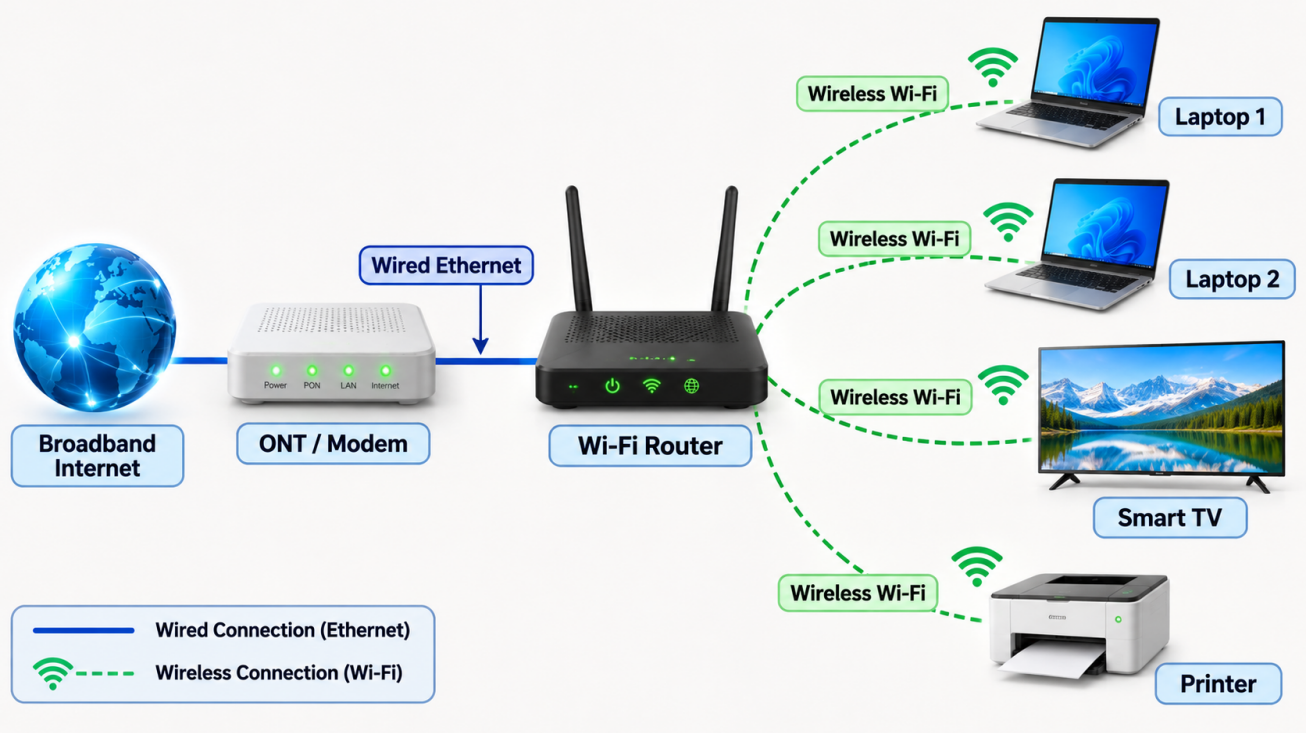
SOHO NETWORK ASSIGNMENT

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hardware Component	Brand / Model Suggestion	Purpose
Broadband Internet connection	Airtel Xstream Fiber / JioFiber / ACT Fibernet	Provides Internet access through the service provider.
ONT / Modem	ISP-provided fibre ONT or modem	Receives the broadband connection and provides an Ethernet connection to the router.
Wi-Fi Router	TP-Link Archer C6 (AC1200)	Shares the Internet connection and provides Wi-Fi to the devices.
Laptops (2)	Lenovo IdeaPad / HP Pavilion / Acer Aspire	Connect wirelessly for browsing and office work.
Smart TV	Samsung / LG / Xiaomi Wi-Fi Smart TV	Connects to Wi-Fi for streaming and online services.
Printer	Epson EcoTank L3250 Wi-Fi / HP DeskJet 2720	Allows printing over the local network.
Ethernet cable (Cat6)	D-Link / UGREEN Cat6 RJ45 cable (1–3 m)	Connects the ONT/modem to the router; can also connect a wired printer.
Optional Wi-Fi extender	TP-Link RE305 / RE200	Extends Wi-Fi coverage if the signal is weak in another room.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Step	Instruction
1	Connect the broadband ONT/modem to the router's WAN/Internet port using an Ethernet cable. Switch on both devices.
2	Connect a laptop to the router using Wi-Fi or an Ethernet cable and open a web browser.
3	Enter the router's address in the browser, commonly 192.168.0.1, 192.168.1.1 or tplinkwifi.net.
4	Log in using the details supplied with the router. Open Administration or System Settings and change the default admin password to a strong, unique password.
5	Open Wireless Settings and change the Wi-Fi network name (SSID) to a unique name, for example SOHO_Network.
6	Open Wireless Security and select WPA2-Personal / WPA2-PSK (AES). Set a strong Wi-Fi password.
7	Make sure DHCP is enabled so the laptops can automatically receive IP addresses from the router.
8	Save or apply the settings. Restart the router if required, then reconnect all devices using the new SSID and password.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

The network will use the private IP range 192.168.1.0/24. The router will use 192.168.1.1 as its address and default gateway. The printer and Smart TV will use static IP addresses, while the laptops will receive addresses automatically through DHCP.

Device	MAC Address (Example)	IP Address	Assignment Type
Wi-Fi Router	02:11:22:33:44:01	192.168.1.1	Gateway
Printer	02:11:22:33:44:10	192.168.1.20	Static
Smart TV	02:11:22:33:44:20	192.168.1.30	Static
Laptop 1	02:11:22:33:44:31	192.168.1.101	DHCP
Laptop 2	02:11:22:33:44:32	192.168.1.102	DHCP

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible Cause	**How to Diagnose**	**Steps to Fix**
1. Incorrect Wi-Fi name or password	Check whether the laptop is connected to the correct SSID. Look for an 'incorrect password' or authentication error. Check whether other devices can connect.	Select the correct Wi-Fi network. Forget the saved network and reconnect using the correct password.
2. Wi-Fi adapter disabled or driver issue	Check whether Wi-Fi or Airplane mode is switched off. On Windows, check Device Manager for an error beside the wireless network adapter.	Turn on Wi-Fi and switch off Airplane mode. Restart the laptop and update or reinstall the wireless adapter driver if needed.
3. IP address or DHCP problem	Open Command Prompt and run ipconfig. An address beginning with 169.254 may mean the laptop did not receive an IP address from the router.	Set IPv4 to obtain an IP address automatically. Run ipconfig /release and then ipconfig /renew to request a new address.
4. Weak Wi-Fi signal	Check the Wi-Fi signal strength and try connecting near the router. If the connection improves nearby, distance or walls may be affecting the signal.	Move closer to the router or place it in a more central, open location. Use the 2.4 GHz band for better range if available.
5. Router, modem or ISP problem	Check whether the second laptop and Smart TV can access the Internet. If no device can connect, inspect the router/ONT status lights and Internet status.	Restart the modem/ONT and router. If the Internet connection does not return, contact the ISP to check for an outage or line problem.